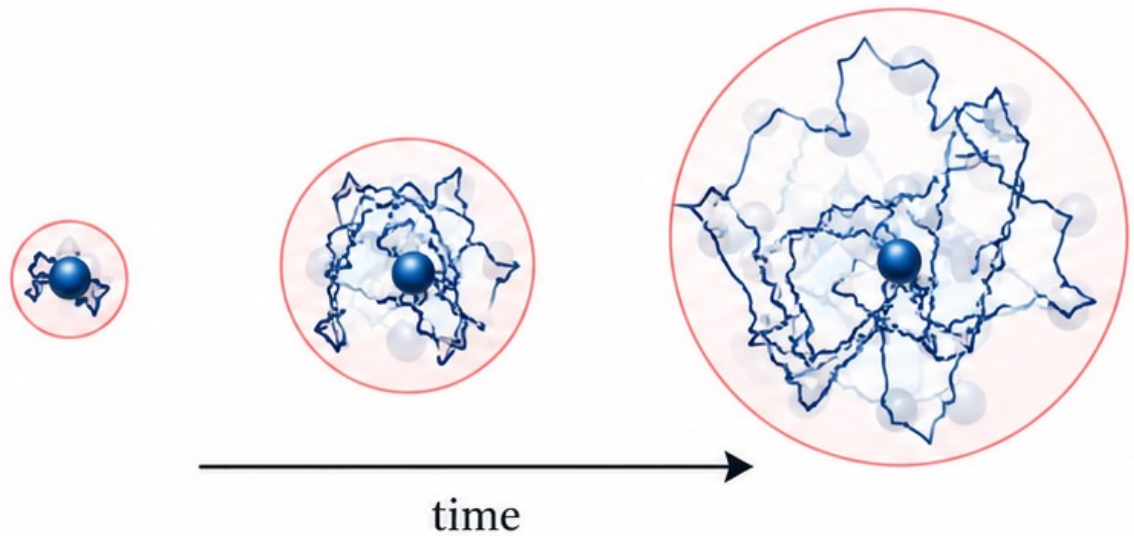
MAT 504

Computer Modelling in Materials Science

Week 3 — Fundamentals of Molecular Dynamics

Assoc. Prof. Betül Uralcan
Sabancı University

Case study: diffusion from continuum to atomistic description



Continuum and atomistic perspectives

Fundamentals of Molecular Dynamics

Fundamentals of Molecular Dynamics

- Carry out atomistic simulations of various processes
- Analyze atomistic simulations
- Advanced molecular dynamics
- Visualize atomistic/molecular data
- Understand how to link atomistic simulation results with continuum models within a multi-scale scheme

In today's class

Case Study: Diffusion

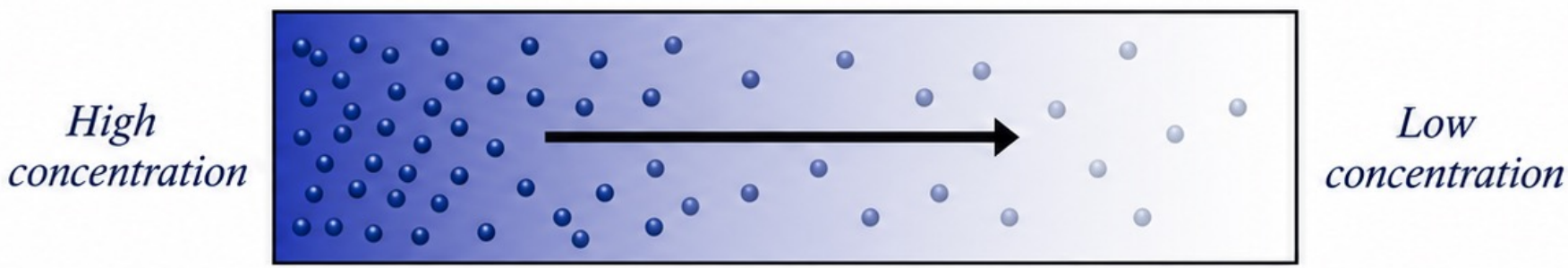
- Macroscopic perspective (continuum model)
- Microscopic Perspective (atomistic model)

Goals of today's lecture:

- Introduce diffusion through a continuum model and an atomistic model
- Develop appreciation for distinction of continuum and atomistic approach
- Develop equations/models for diffusion problem from both perspectives
- Develop atomistic simulation approach (e.g. algorithm, pseudocode, etc.) and apply to describe diffusion (calculate diffusivity)

Introduction: Diffusion

- Particles move from a domain with **high concentration** to an area of **low concentration**
- **Macroscopically**, diffusion measured by change in concentration
- **Microscopically**, diffusion is process of spontaneous net movement of particles
  - Result of random motion of particles ("Brownian motion")



$$c = m/V = c(\vec{x}, t)$$

Ink droplet in water



Relevant scales in materials

λ(atom) << ξ(crystal) << d(grain) << dx, dy, dz << H, W, D

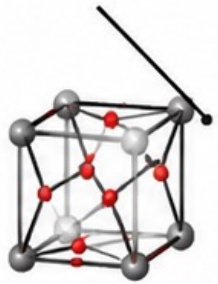
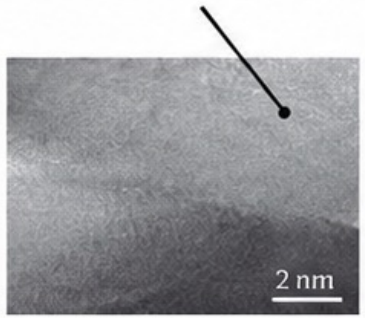


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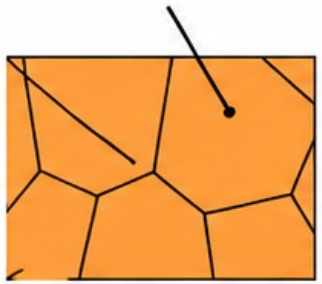


Fig. 8.7 in: Buehler, M. *Atomistic Modeling of Materials Failures*. Springer, 2008.
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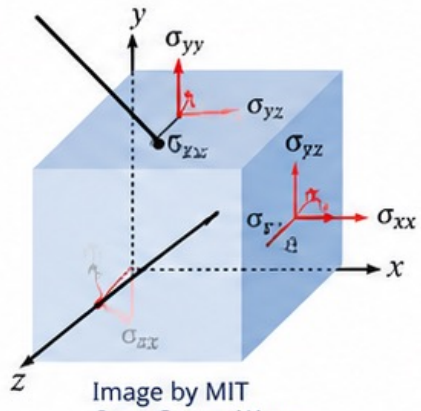


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Bottom-up → ← **Top-down**

Atomic viewpoint:

- Explicitly consider discrete atomistic structure
- Solve for atomic trajectories and infer from these about material properties & behavior
- Features internal length scales (atomic distance)

"Many-particle system with statistical properties"

Relevant scales in materials

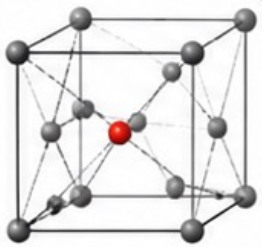
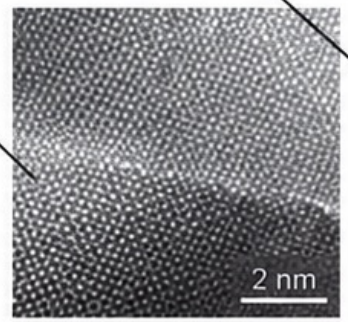


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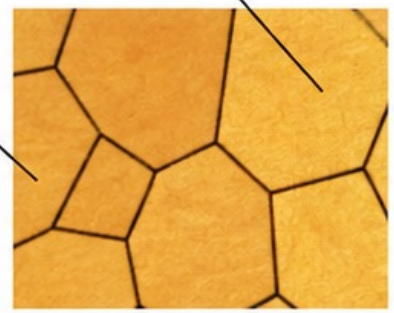


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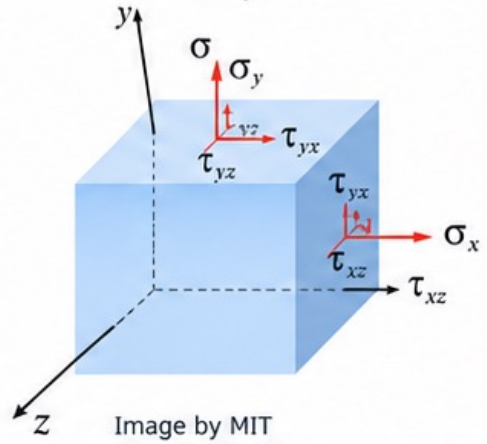


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Top-down

Continuum viewpoint:

- Treat material as matter with no internal structure.
- Develop mathematical model (governing equation) based on representative volume element (RVE), contains "enough" material such that internal structure can be neglected.
- Features no characteristic length scales, provided RVE is large enough.

"PDE with parameters"

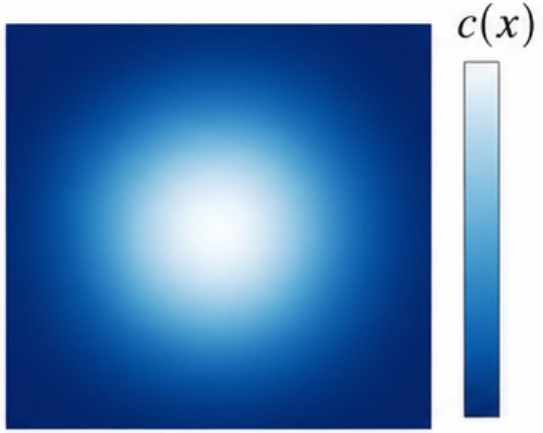
Goal of this section

Goal of this section

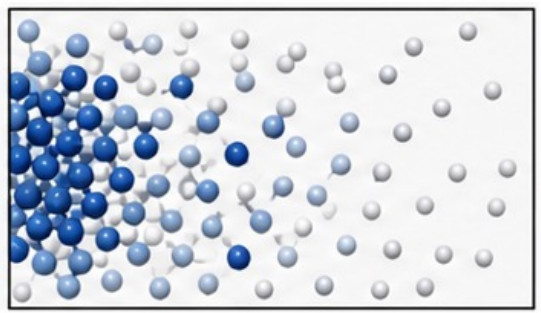
- **Diffusion** as example.
- **Continuum description** (top-down approach), partial differential equation.
- **Atomistic description** (bottom-up approach), based on dynamics of molecules, obtained via numerical simulation of the molecular dynamics.

Continuum description
(top-down approach)

$$\frac{\partial c(x,t)}{\partial t} = D\nabla^2 c(x,t)$$



Atomistic description
(bottom-up approach)



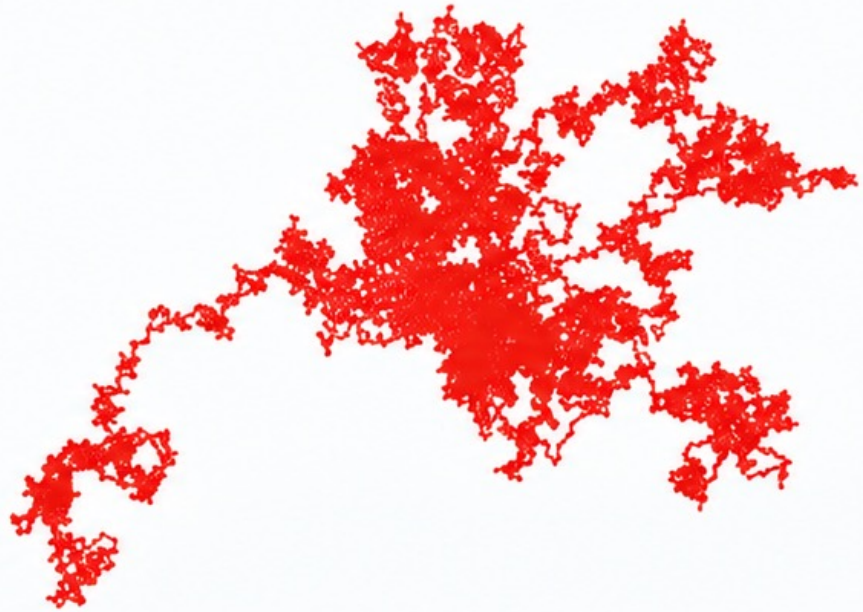
Observation of Diffusion Properties

Observation of Diffusion Properties

Continuum and **Atomistic** Modeling

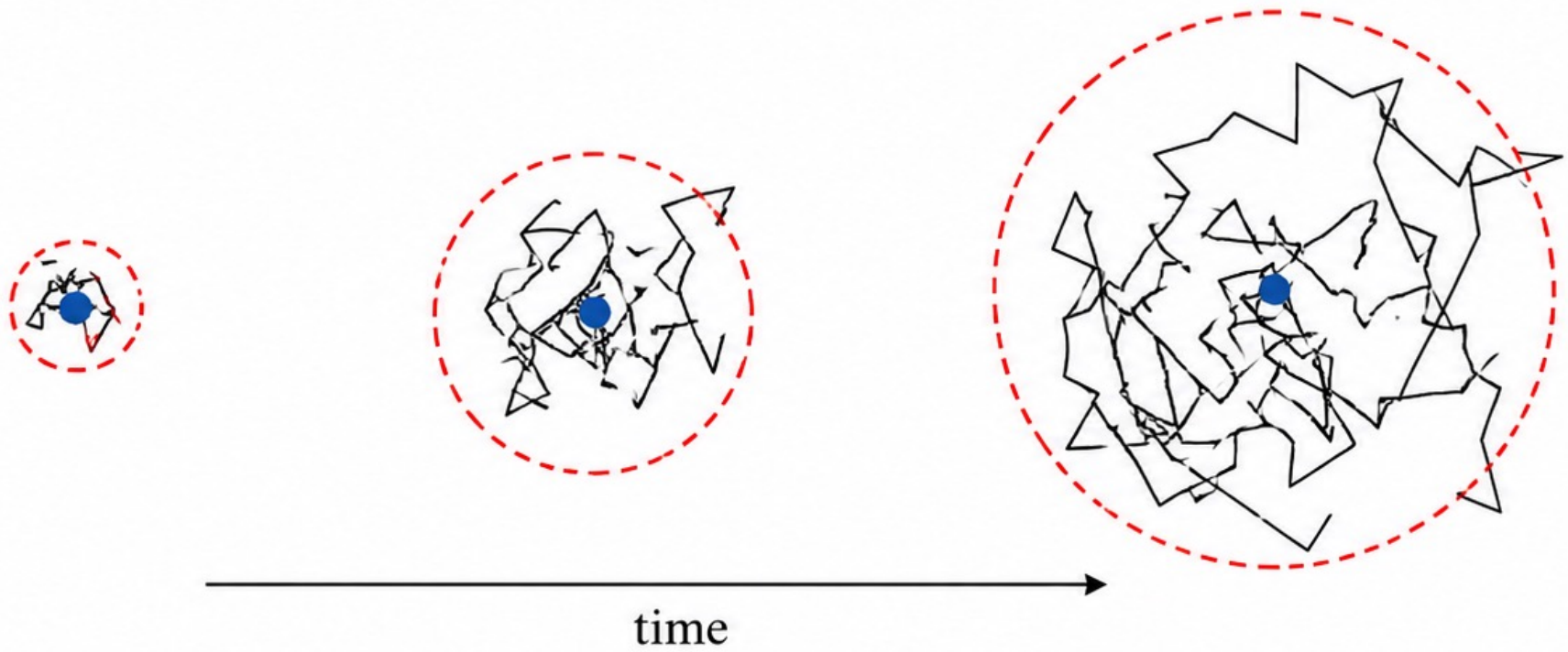
Microscopic mechanism: random walk or Brownian motion

- Brownian motion was first observed (1827) by the British botanist Robert Brown (1773-1858) when studying pollen grains in water.
- Initially thought to be sign of life, but later confirmed to be also present in inorganic particles.
- The effect was finally explained in 1905 by Albert Einstein, who realized it was caused by water molecules randomly smacking into the particles.



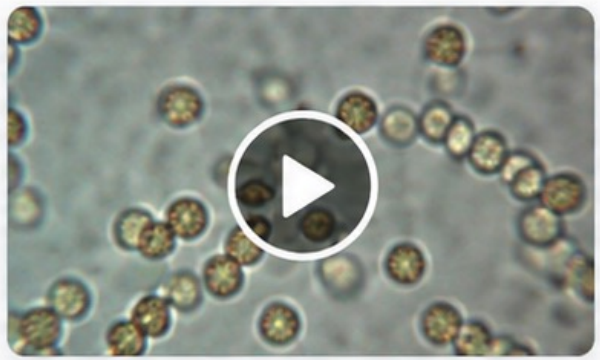
Public domain image.

Brownian motion leads to net particle movement



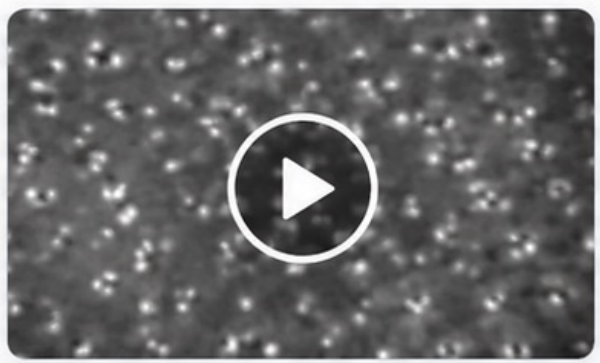
Particle slowly moves away from its initial position.

Simulation of Brownian motion



Pollen Grains in Water

https://www.youtube.com/watch?v=R5t-oA796to



Brownian motion demonstration

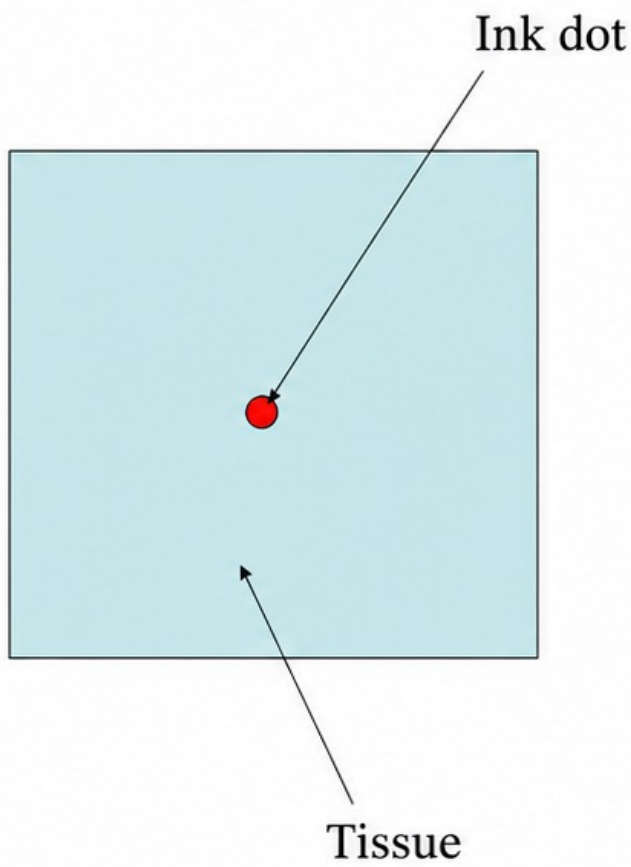
https://www.youtube.com/watch?v=gPMVaAnij88

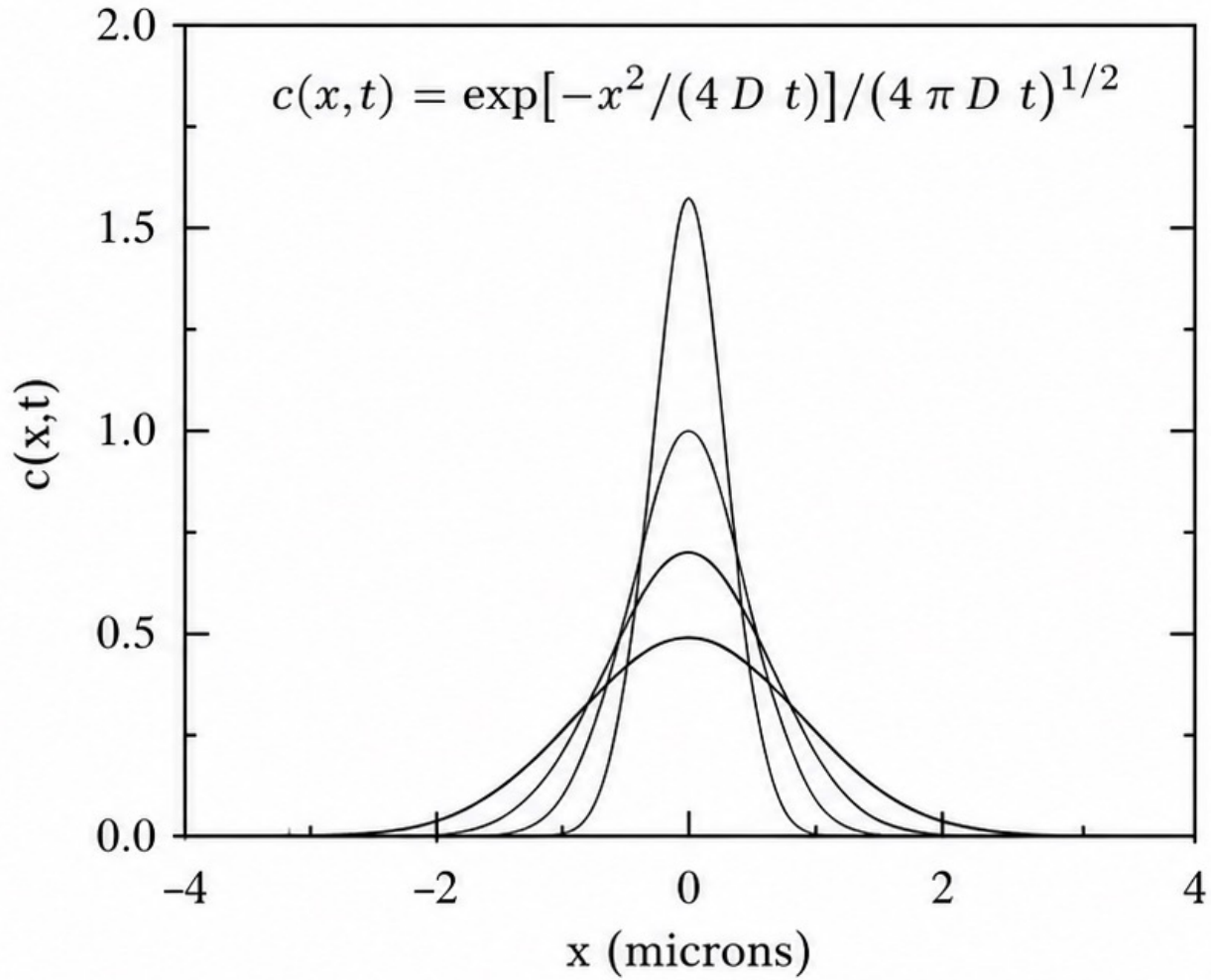
Observation of Diffusion Properties

Observation of Diffusion Properties

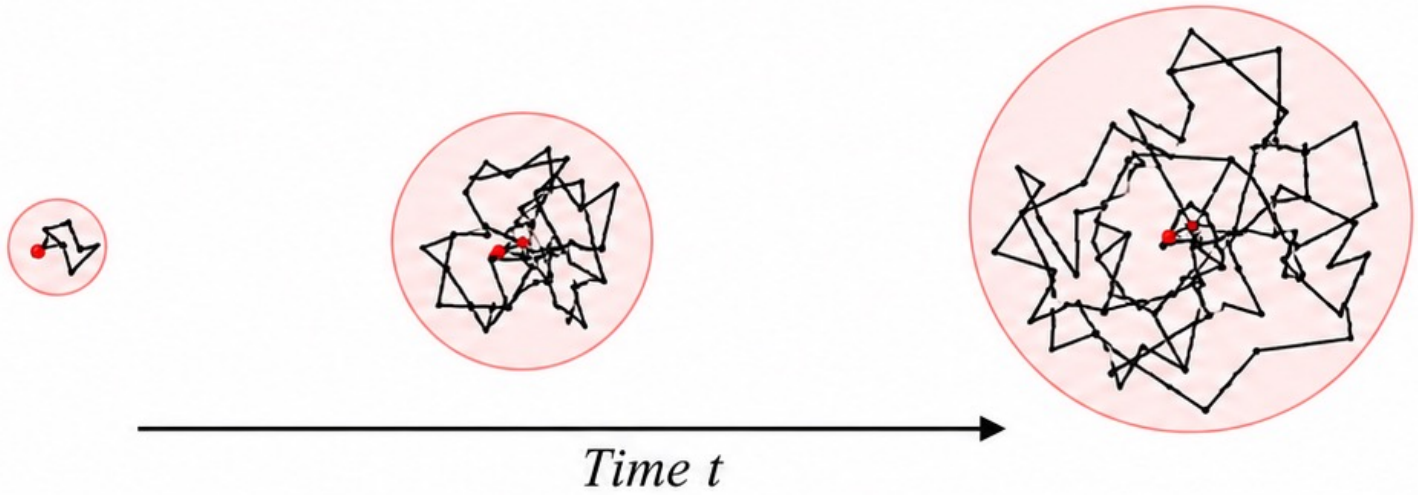
Continuum and Atomistic Modeling

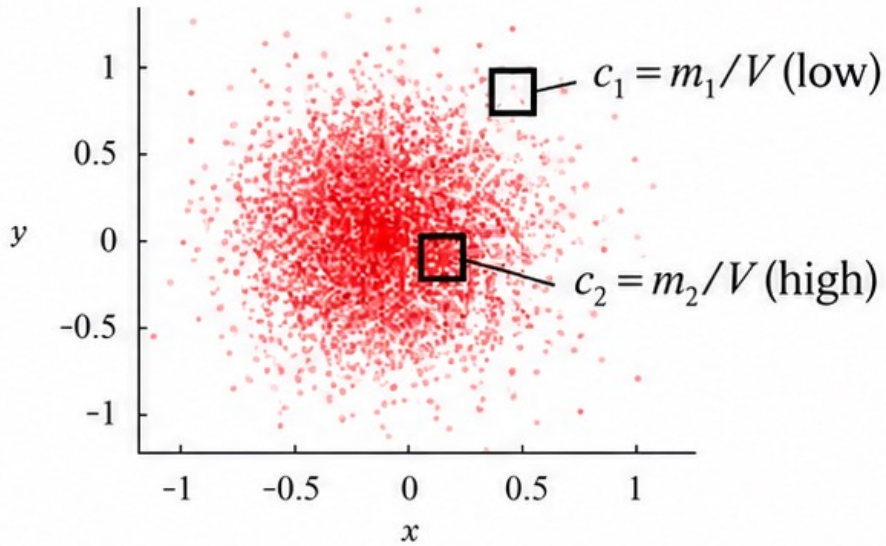
Macroscopic observation: concentration change

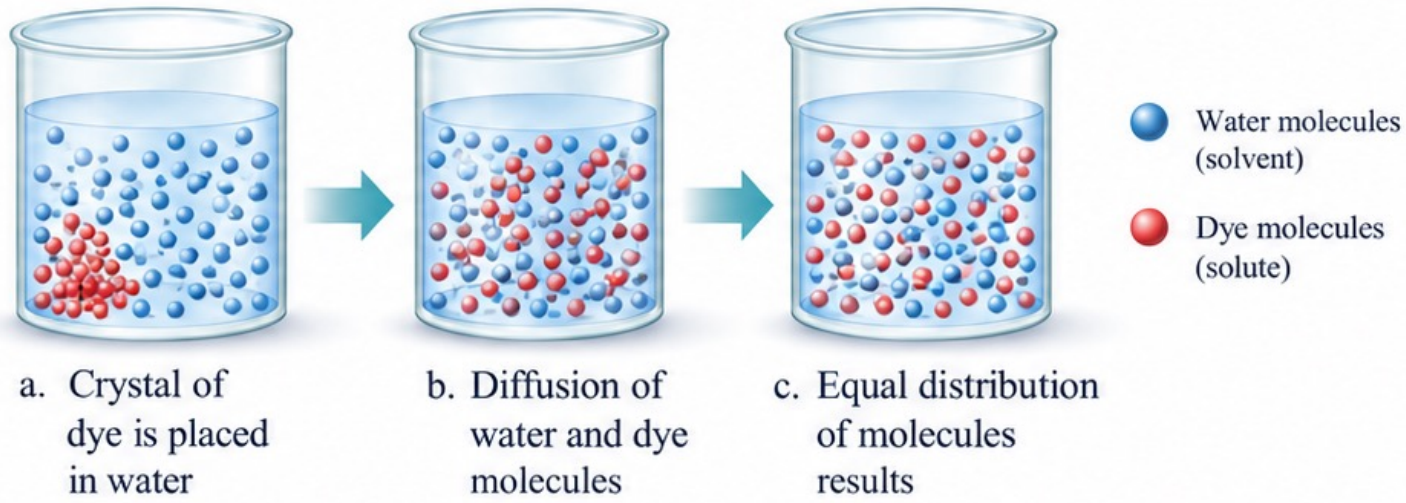




Brownian motion leads to net particle movement





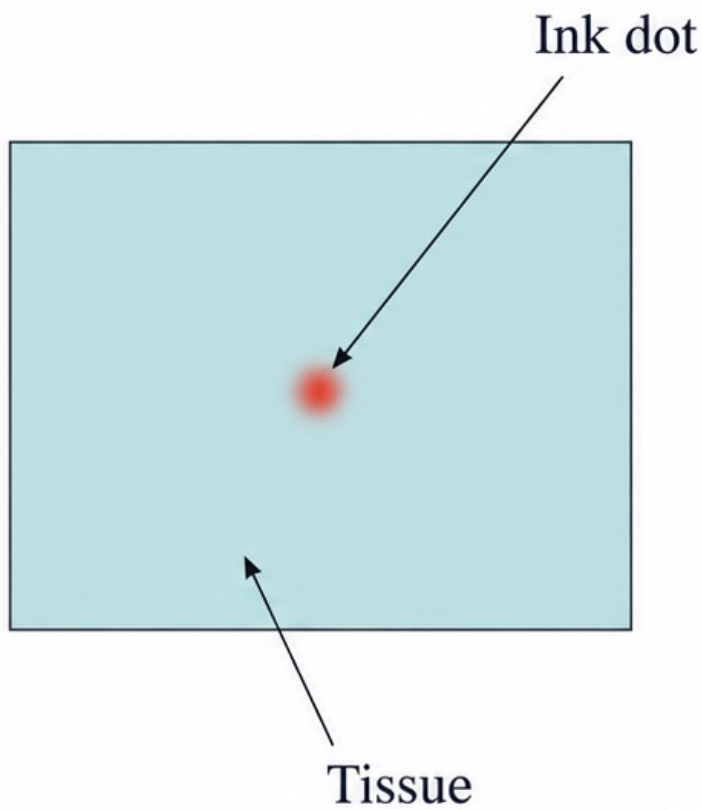


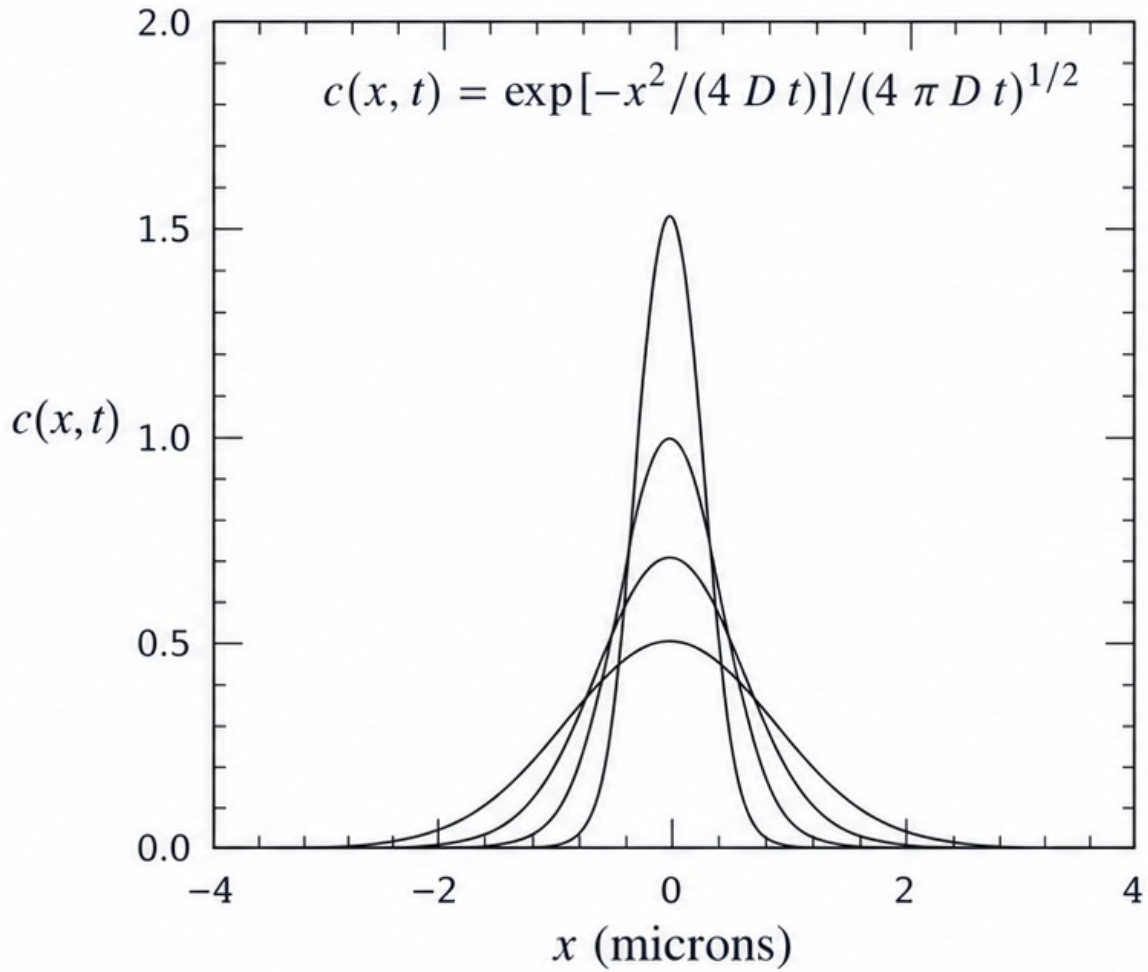
a. Crystal of dye is placed in water

b. Diffusion of water and dye molecules

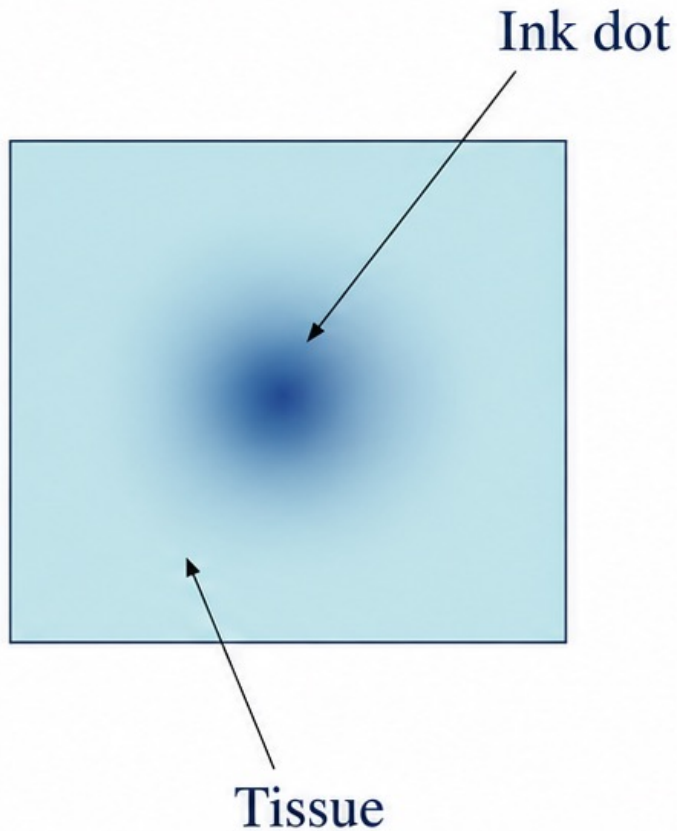
c. Equal distribution of molecules results

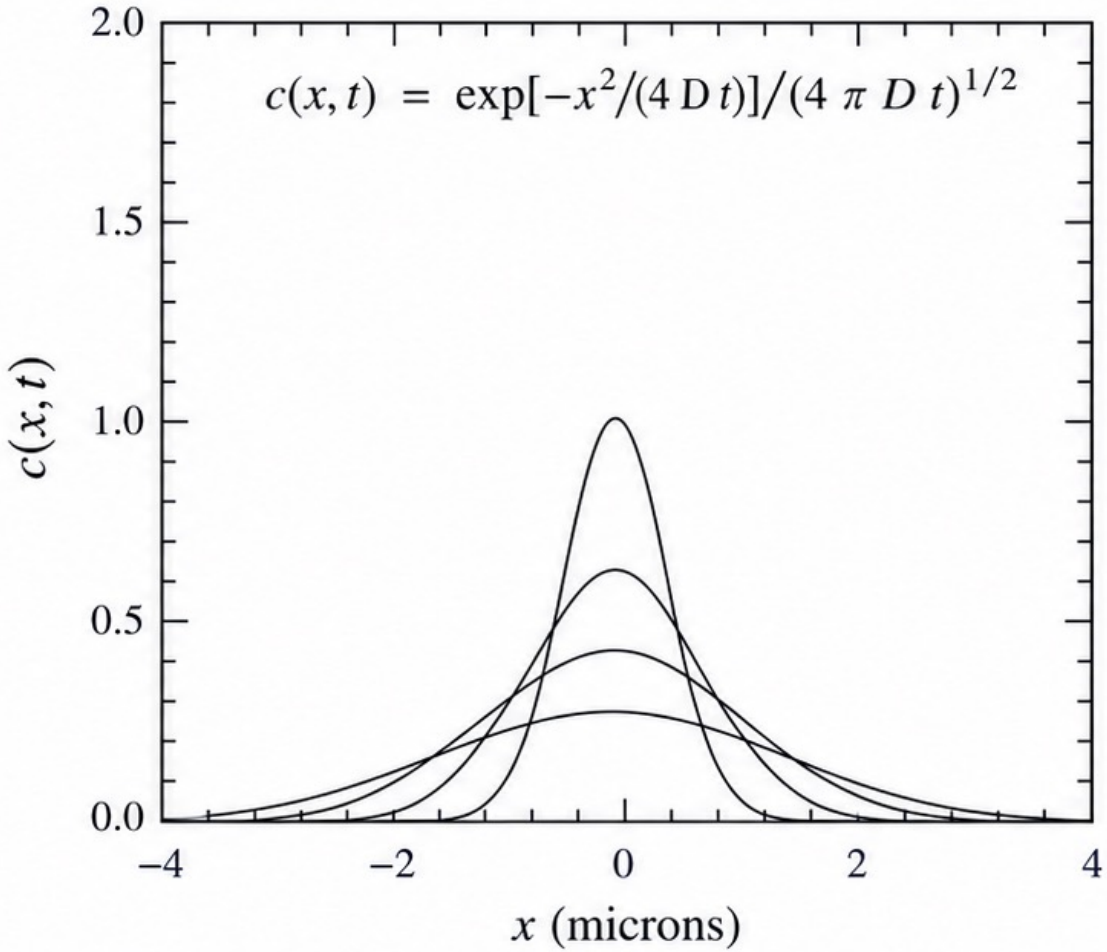
Macroscopic observation: concentration change



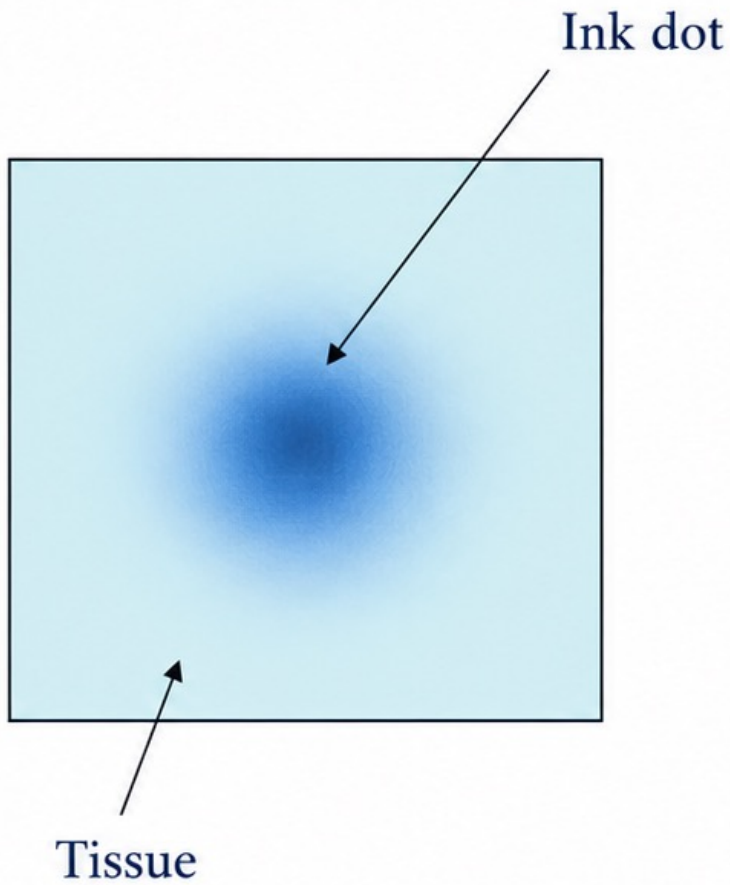


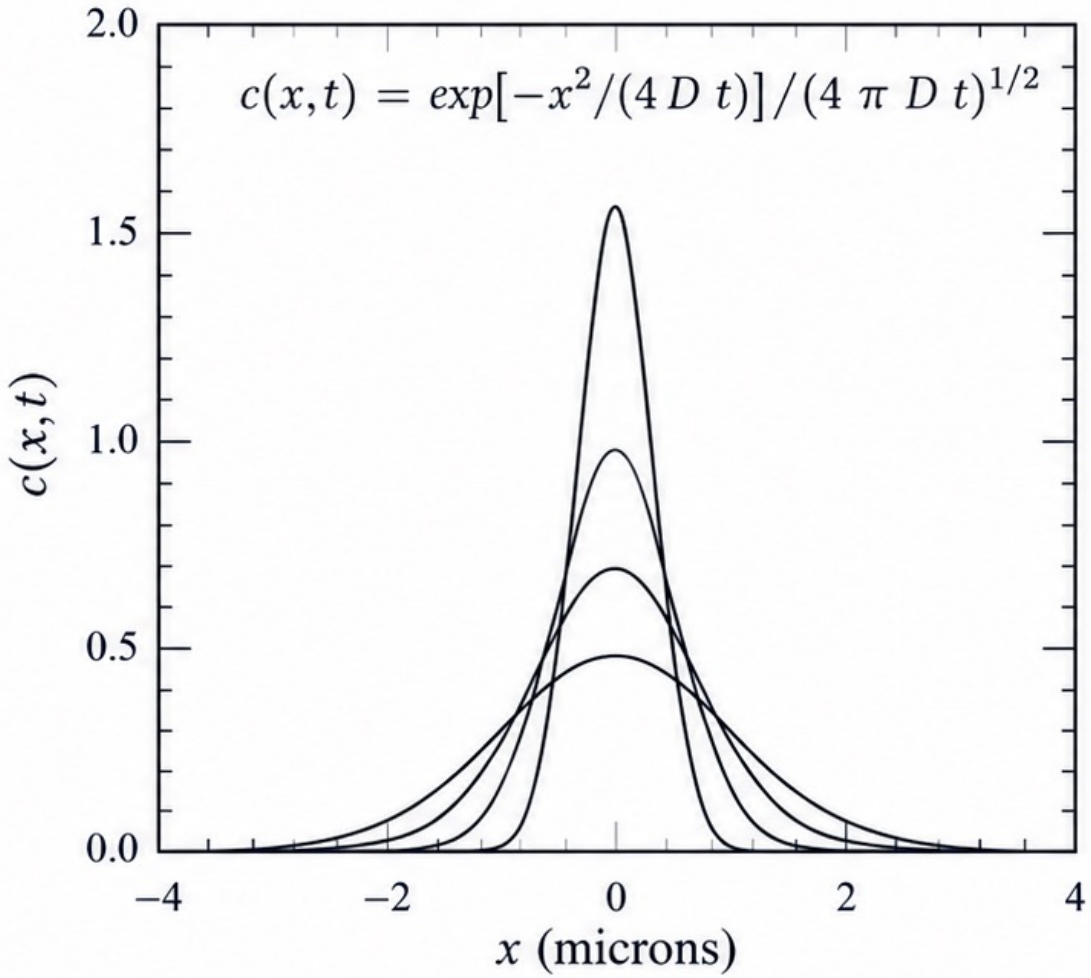
Macroscopic observation: concentration change



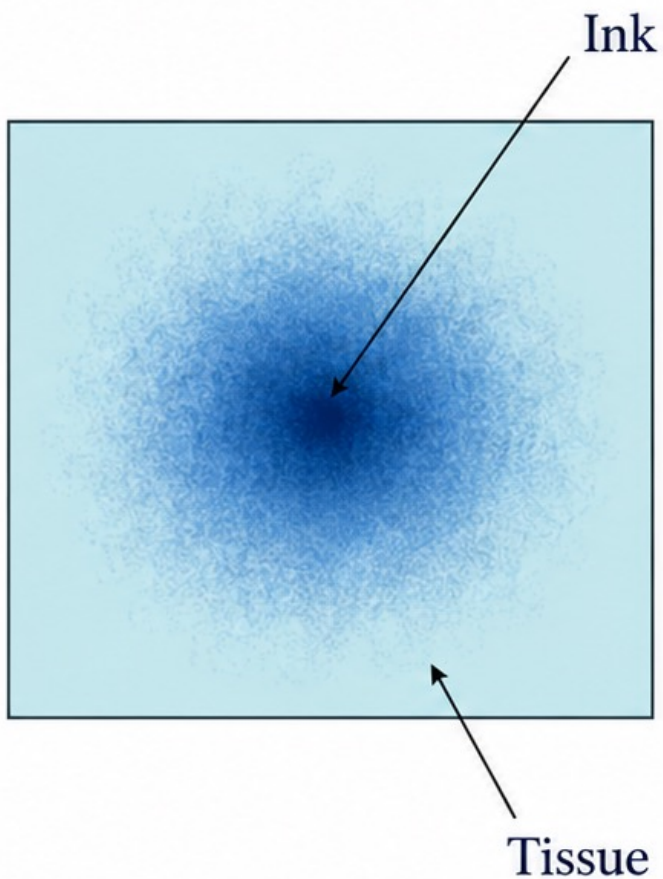


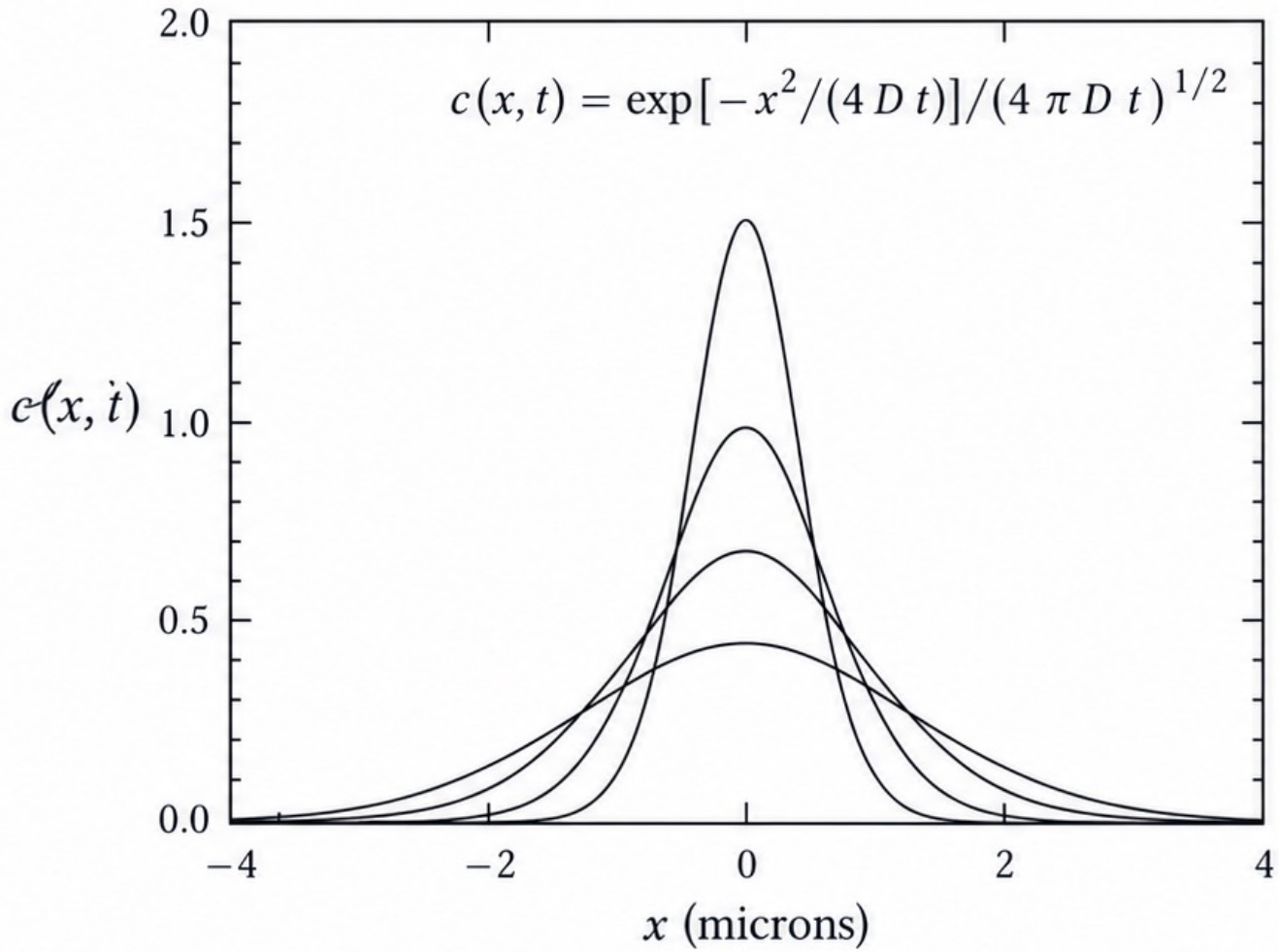
Macroscopic observation: concentration change





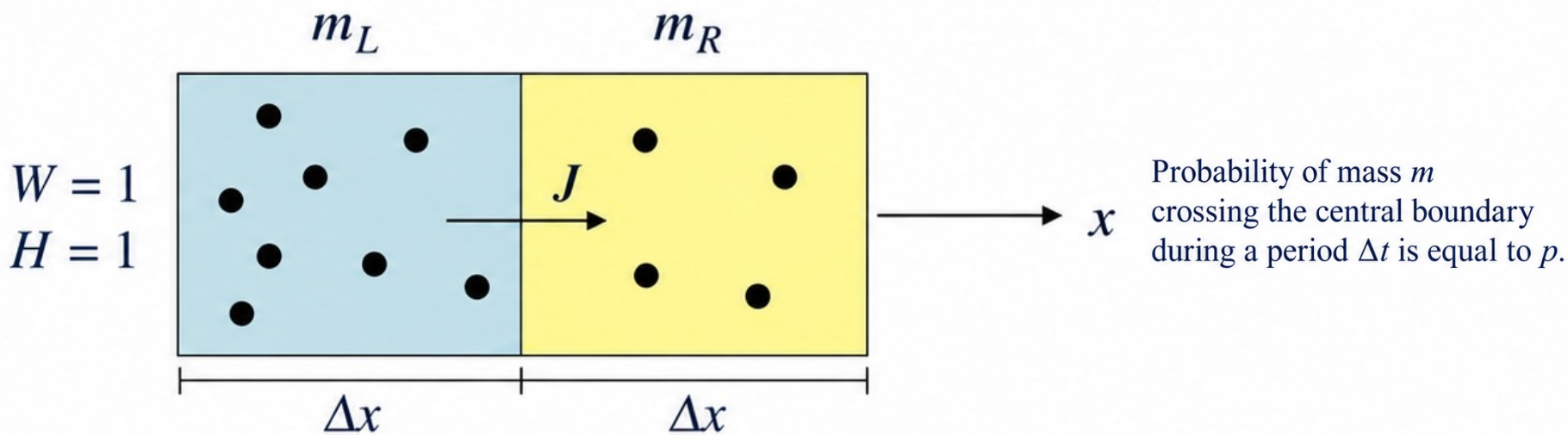
Macroscopic observation: concentration change





Approach 1: Continuum model

- Develop differential equation based on differential element

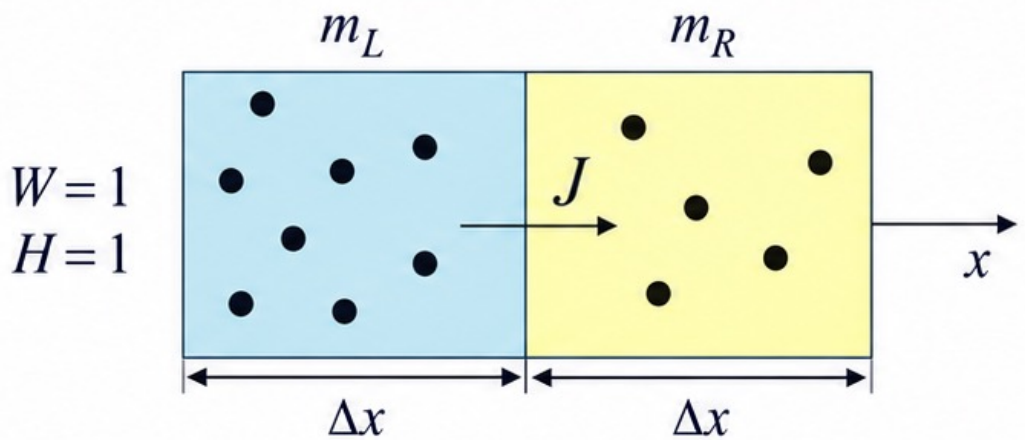


$\boldsymbol{J}$: Mass flux (mass per unit time per unit area)

Concept: Balance mass in a differential volume element; much greater in dimension than inhomogeneities ("sufficiently large REV")

Approach 1: Continuum model

- Develop differential equation based on differential element



Probability of mass m crossing the central boundary during a period Δt is equal to p.

$$J_L = \frac{1}{1 \times 1} \frac{p}{\Delta t} m_L$$ Mass flux from left to right

$$J_R = \frac{1}{1 \times 1} \frac{p}{\Delta t} m_R$$ Mass flux from right to left

Effective mass flux

$$J = \frac{1}{1 \times 1} \frac{p}{\Delta t} (m_L - m_R)$$

- *More mass, more flux (m_L is ~ to number of particles).*

Continuum model of diffusion

Express in terms of mass concentrations

$$c = \frac{m}{V} \qquad m = cV$$

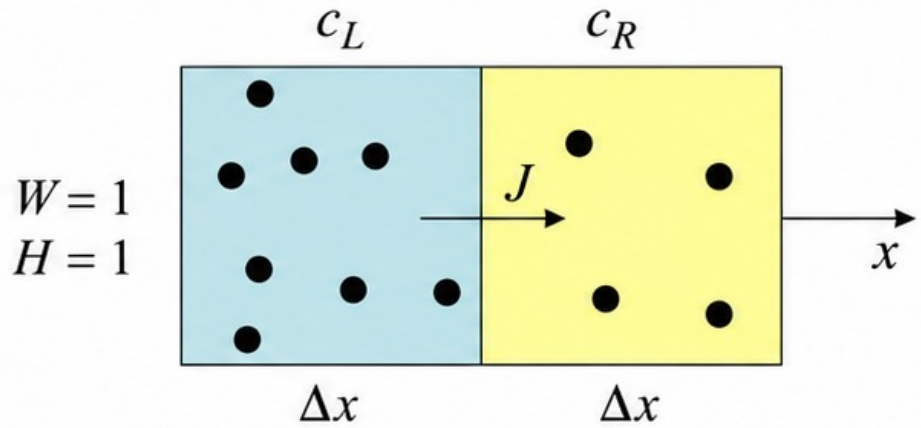
$$J = \frac{P}{\Delta t}(m_L - m_R)$$

$$J = \frac{1}{1 \times 1}\frac{P}{\Delta t}(c_L V - c_R V)$$

$$= \frac{P}{\Delta t}(c_L - c_R)\,\Delta x \times 1$$

$$= \frac{P\,\Delta x}{\Delta t}(c_L - c_R)$$

$$\Delta c = c_L - c_R$$



Flux in terms of concentration difference

$$J = \frac{P\,\Delta x}{\Delta t}(c_L - c_R) = \underbrace{-\frac{P\,\Delta x}{\Delta t}}_{D}(c_R - c_L) = -D\frac{\Delta c}{\Delta x}$$

Continuum (gradient) form

$$J = -D\frac{\partial c}{\partial x}$$

The flux is proportional to the (negative) concentration gradient.

Diffusion constant and Fick's first law

From mass balance to flux

Express in terms of mass concenstrations

$$c = \frac{m}{V}$$

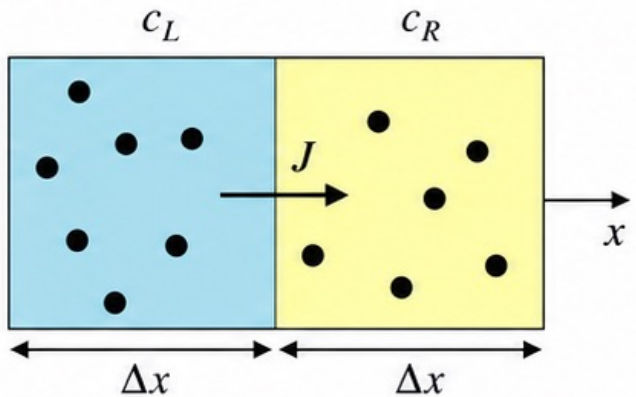
$$J = \frac{1}{A\,\Delta t}(c_L - c_R)\,\Delta x\,(A)$$

$$= -\frac{\Delta c}{\Delta t}\,V$$

$$J = -\frac{\Delta c}{\Delta t}\Delta x = -\frac{\Delta c}{\Delta x}\,\frac{\Delta x^2}{\Delta t}$$

Concentration gradient

$W = 1$
$H = 1$



$$J = -D\,\frac{dc}{dx}$$

$$D = \frac{(\Delta x)^2}{\Delta t}$$

Diffusion constant
measures how much mass moves per unit time (in square of distance per unit time)

Adolf Fick (1829–1901)



Adolf Fick introduced empirical laws of diffusion (1855), relating mass flux to concentration gradients.

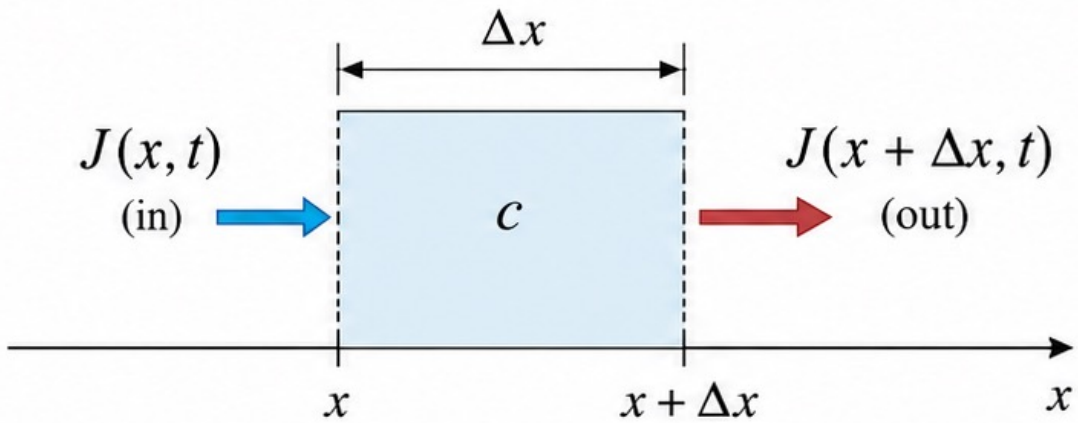
- The diffusion constant D describes how much mass moves per unit time.
- D is related to movement over a distance Δx in a time Δt by $D = (\Delta x)^2/\Delta t$.
- Fick's first law: the mass flux is proportional to (and opposite to) the concentration gradient.

Fick's second law: time dependence

Fick's first law

$$J = -D\frac{dc}{dx}$$

J: diffusion flux (mass per area per time)
c: concentration (mass per volume)
D: diffusion constant



Change in concentration due to flux difference

Net influx into the control volume = incoming flux − outgoing flux

$$\Delta M = [J(x,t) - J(x+\Delta x,t)]\,A\Delta t$$

Change in concentration (mass/volume) is

$$\Delta c = \frac{\Delta M}{A\Delta x} = -\frac{J(x+\Delta x,t) - J(x,t)}{\Delta x}\,\Delta t$$

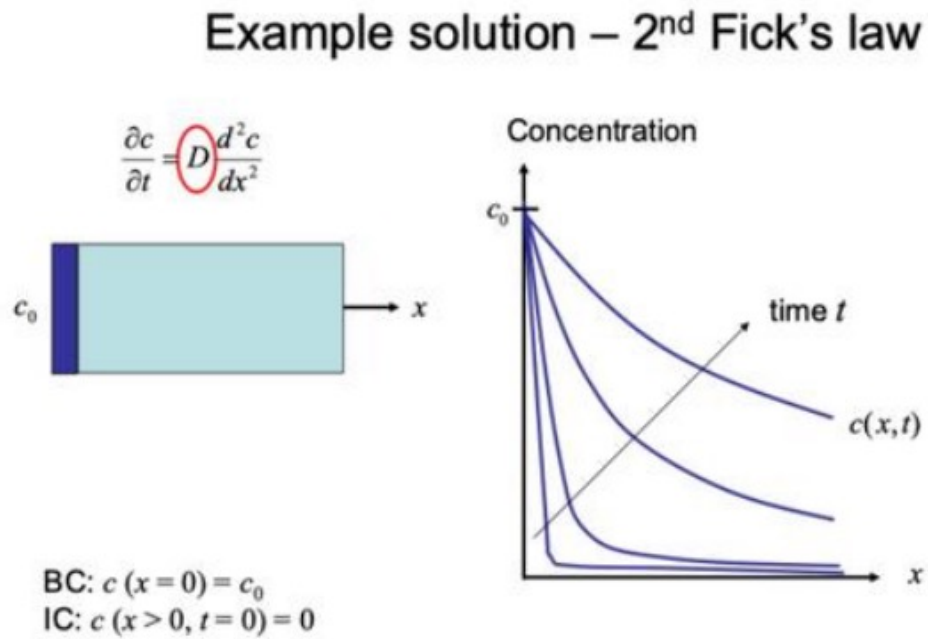
Take the limit $\Delta x,\ \Delta t \to 0$ and use Fick's first law:

$$\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2}$$

Fick's second law
(time dependence of concentration)

- If more mass flows in than out, the concentration in the element increases ($\partial c/\partial t > 0$).
- If more mass flows out than in, the concentration decreases ($\partial c/\partial t < 0$).
- Fick's second law describes how concentration evolves with time due to spatial variations in concentration (curvature through $\partial^2 c/\partial x^2$).

Example solution: Fick's second law



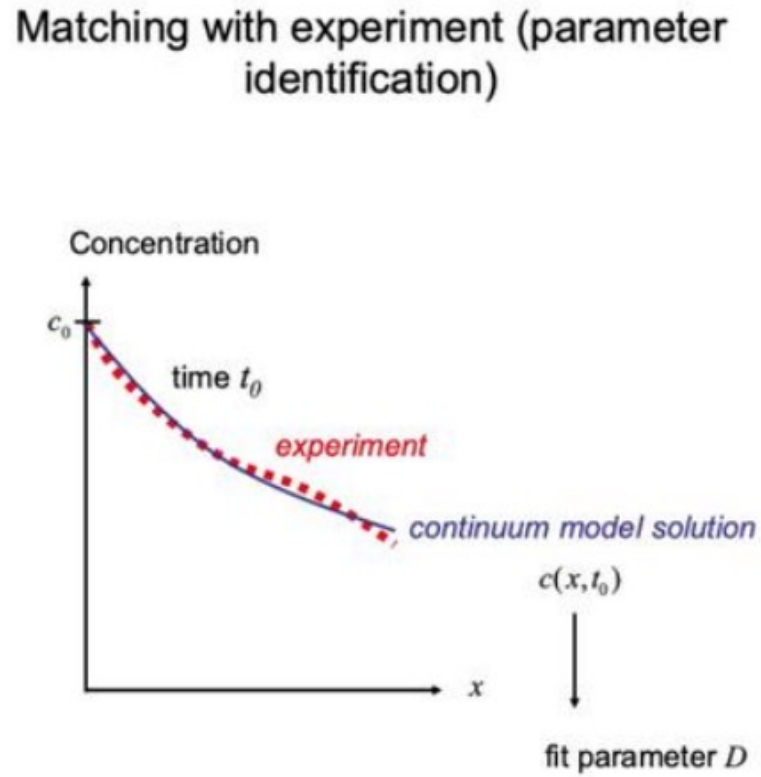
Need diffusion coefficient to solve for distribution!

How to obtain the diffusion coefficient?

How to obtain diffusion coefficient?

- Laboratory experiment
- Study distribution of concentrations (previous slide)
- Then “fit” the appropriate diffusion coefficient so that the solution matches
- Approach can then be used to solve for more complex geometries, situations etc. for which no lab experiment exists
- “Top down approach”

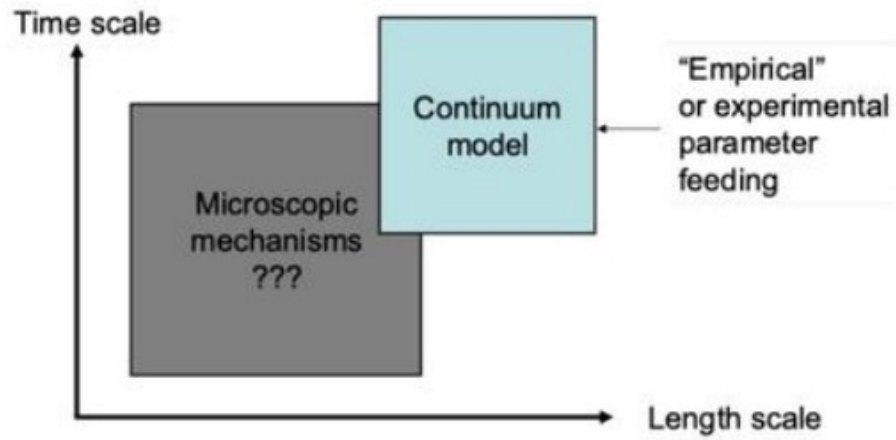
Matching with experiment



Summary

Summary

- Continuum model requires parameter that describes microscopic processes inside the material
- Typically need experimental measurements to calibrate



Approach 2: Atomistic model

Approach 2: Atomistic model

- Atomistic model provides an alternative approach to describe diffusion
- Enables us to directly calculate the diffusion constant from the trajectory of atoms ("microscopic definition")
- Approach: Consider set of atoms/molecules



- Follow their trajectory and calculate how fast atoms leave their initial position

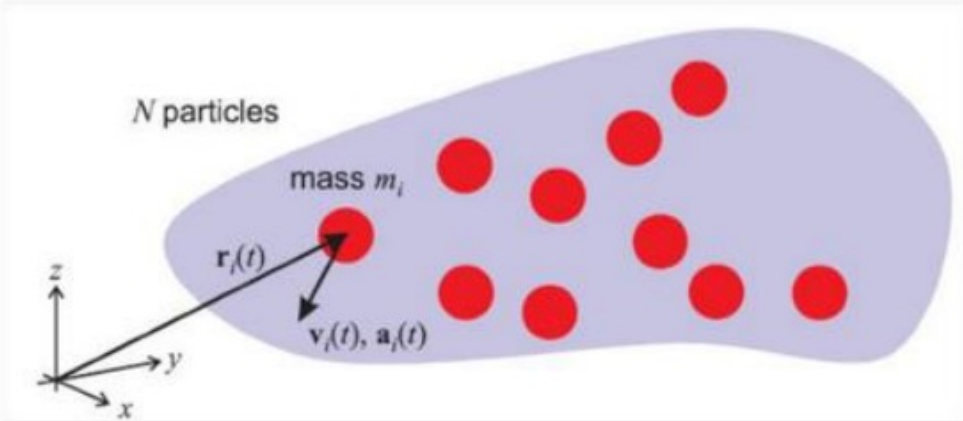
Follow this quantity over time

$$D = p\frac{\Delta x^2}{\Delta t}$$

Recall: Diffusion constant relates to the "ability" of particle to move a distance Δx^2 over a time Δt

Molecular dynamics: simulate trajectories of atoms

Molecular dynamics – simulate trajectory of atoms



Goal: Need an algorithm to predict positions, velocities, accelerations as function of time

Solving the equations: what we want

Solving the equations: What we want

To solve those equations: Discretize in time (n steps), Δt time step:

$$r_i(t_0) \rightarrow r_i(t_0 + \Delta t) \rightarrow r_i(t_0 + 2\Delta t) \rightarrow r_i(t_0 + 3\Delta t) \rightarrow \ldots \rightarrow r_i(t_0 + n\Delta t)$$

42

Solving the equations

Solving the equations

To solve those equations: Discretize in time (n steps), Δt time step:

$$r_i(t_0) \rightarrow r_i(t_0 + \Delta t) \rightarrow r_i(t_0 + 2\Delta t) \rightarrow r_i(t_0 + 3\Delta t) \rightarrow \ldots \rightarrow r_i(t_0 + n\Delta t)$$

Recall: Taylor expansion of function f around point a

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f^{(3)}(a)}{3!}(x-a)^3 + \ldots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \ldots.$$

Solving the equations

Solving the equations

To solve those equations: Discretize in time (n steps), Δt time step:

$$r_i(t_0) \rightarrow r_i(t_0+\Delta t) \rightarrow r_i(t_0+2\Delta t) \rightarrow r_i(t_0+3\Delta t) \rightarrow \ldots \rightarrow r_i(t_0+n\Delta t)$$

Recall: Taylor expansion of function f around point a

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f^{(3)}(a)}{3!}(x-a)^3 + \ldots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \ldots.$$

Taylor series expansion $r_i(t)$ around $a = t_0$ $x = t_0 + \Delta t$

$$x - a = t_0 + \Delta t - t_0 = \Delta t$$

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Taylor series expansion $r_i(t)$ around $a = t_0$ $x = t_0 + \Delta t$

$$x - a = t_0 + \Delta t - t_0 = \Delta t$$

$$r_i(t_0+\Delta t) = r_i(t_0) + v_i(t_0)\Delta t + \frac{1}{2}a_i(t_0)\Delta t^2 + \ldots$$

Forward expansion

Taylor expansion of x(t)

Taylor expansion of $r_i(t)$

$$r_i(t_0+\Delta t) = r_i(t_0) + v_i(t_0)\Delta t + \frac{1}{2}a_i(t_0)\Delta t^2 + \ldots$$

$$r_i(t_0-\Delta t) = r_i(t_0) - v_i(t_0)\Delta t + \frac{1}{2}a_i(t_0)\Delta t^2 + \ldots$$

$$a = t_0 \qquad x = t_0 + \Delta t \qquad x - a = t_0 + \Delta t - t_0 = \Delta t$$

$$a = t_0 \qquad x = t_0 - \Delta t \qquad x - a = t_0 - \Delta t - t_0 = -\Delta t$$

Taylor expansion of x(t)

Taylor expansion of $r_i(t)$

$$r_i(t_0+\Delta t) = r_i(t_0) + v_i(t_0)\Delta t + \frac{1}{2}a_i(t_0)\Delta t^2 + \ldots$$ *Forward expansion*

$$+\left[\, r_i(t_0-\Delta t) = r_i(t_0) - v_i(t_0)\Delta t + \frac{1}{2}a_i(t_0)\Delta t^2 + \ldots \,\right]$$ *Backward expansion*

$$\downarrow$$

$$r_i(t_0-\Delta t) + r_i(t_0+\Delta t) = 2r_i(t_0) - v_i(t_0)\Delta t + v_i(t_0)\Delta t + a_i(t_0)\Delta t^2 + \ldots$$

Taylor expansion of x(t)

Taylor expansion of $r_i(t)$

$$r_i(t_0+\Delta t) = r_i(t_0) + v_i(t_0)\Delta t + \frac{1}{2}a_i(t_0)\Delta t^2 + \ldots$$

$$+\left[\, r_i(t_0-\Delta t) = r_i(t_0) - v_i(t_0)\Delta t + \frac{1}{2}a_i(t_0)\Delta t^2 + \ldots \,\right]$$

$$\downarrow$$

$$r_i(t_0-\Delta t) + r_i(t_0+\Delta t) = 2r_i(t_0) - \cancel{v_i(t_0)\Delta t + v_i(t_0)\Delta t} + a_i(t_0)\Delta t^2 + \ldots$$

$$\downarrow$$

$$r_i(t_0+\Delta t) = 2r_i(t_0) - r_i(t_0-\Delta t) + a_i(t_0)\Delta t^2 + \ldots$$

Taylor expansion of x(t)

Taylor expansion of $r_i(t)$

$$r_i(t_0+\Delta t) = r_i(t_0) + v_i(t_0)\Delta t + \frac{1}{2}a_i(t_0)\Delta t^2 + \ldots$$

$$+ \left[r_i(t_0-\Delta t) = r_i(t_0) - v_i(t_0)\Delta t + \frac{1}{2}a_i(t_0)\Delta t^2 + \ldots \right]$$

$$\downarrow$$

$$r_i(t_0-\Delta t) + r_i(t_0+\Delta t) = 2r_i(t_0) - \cancel{v_i(t_0)\Delta t + v_i(t_0)\Delta t} + a_i(t_0)\Delta t^2 + \ldots$$

$$\downarrow$$

$$r_i(t_0+\Delta t) = 2r_i(t_0) - r_i(t_0-\Delta t) + a_i(t_0)\Delta t^2 + \ldots$$

Positions at t_0 | Positions at t_0-Δt | Accelerations at t_0

Verlet Formula

Physics of particle interactions

Physics of particle interactions

Laws of Motion of Isaac Newton (1642 – 1727):

1. Every body continues in its state of rest, or of uniform motion in a right line, unless it is compelled to change that state by forces impressed upon it.
2. The change of motion is proportional to the motive force impresses, and is made in the direction of the right line in which that force is impressed.
3. To every action there is always opposed an equal reaction: or, the mutual action of two bodies upon each other are always equal, and directed to contrary parts.

$$f = m\frac{d^2x}{dt^2} = ma$$

2nd law

Verlet central difference method

Verlet central difference method

$$r_i(t_0 + \Delta t) = \underbrace{2r_i(t_0)}_{\text{Positions at } t_0} - \underbrace{r_i(t_0 - \Delta t)}_{\text{Positions at } t_0-\Delta t} + \underbrace{a_i(t_0)}_{\text{Accelerations at } t_0}\Delta t^2 + \ldots$$

How to obtain accelerations? $f_i = ma_i$, $a_i = f_i / m$

Need forces on atoms!

How are forces calculated?

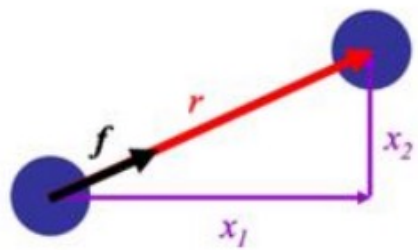
How are forces calculated?

Force magnitude: Derivative of potential energy with respect to atomic distance

$$f = -\frac{\mathrm{d}U(r)}{\mathrm{d}r}$$

To obtain force vector f_i, take projections into the three axial directions

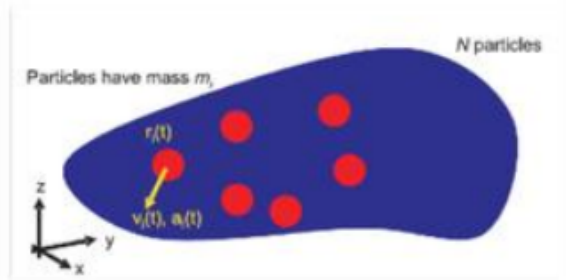
$$f_i = f\frac{x_i}{r}$$



Often: Assume pair-wise interaction between atoms

Molecular dynamics

Molecular dynamics



Follow trajectories of atoms

"Verlet central difference method"

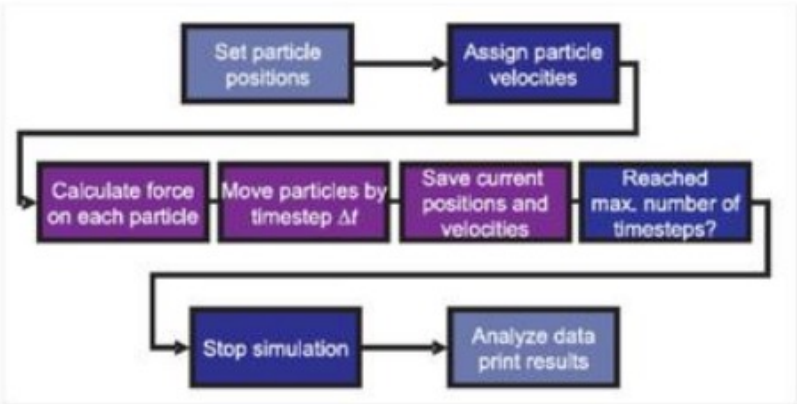
$$r_i(t_0 + \Delta t) = \underbrace{2r_i(t_0)}_{\text{Positions at } t_0} - \underbrace{r_i(t_0 - \Delta t)}_{\text{Positions at } t_0\text{-}\Delta t} + \underbrace{a_i(t_0)}_{\text{Accelerations at } t_0}\Delta t^2 + \ldots$$

$$a_i = f_i / m$$

Summary: atomistic simulation

Summary: Atomistic simulation – numerical approach "molecular dynamics – MD"

- Atomistic model; requires atomistic microstructure and atomic position at beginning
- Step through time by integration scheme
- Repeated force calculation of atomic forces
- Explicit notion of chemical bonds – captured in interatomic potential



Pseudocode

Pseudocode

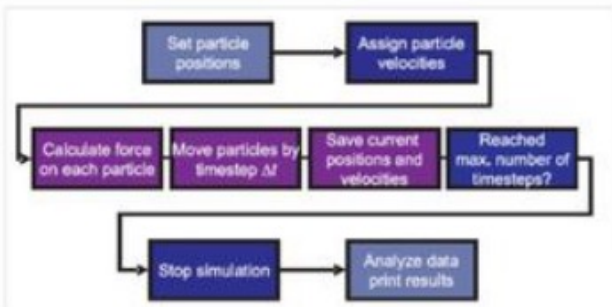
Set particle positions (e.g. crystal lattice)

Assign initial velocities

For (all time steps):
- **Calculate force on each particle (subroutine)**
- **Move particle by time step Δt**
- **Save particle position, velocity, acceleration**

Save results

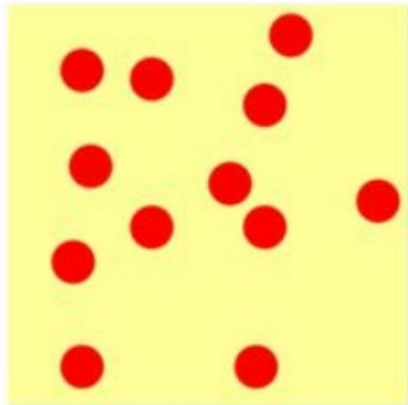
Stop simulation



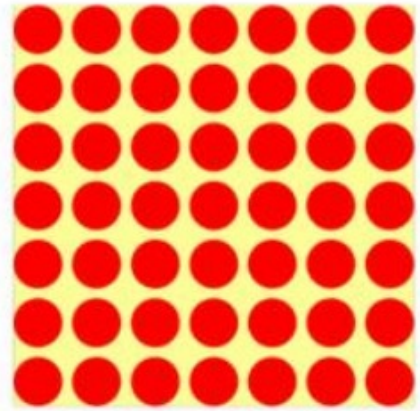
Atomic positions

Atomic positions (initial conditions)

- Typically, have cubical cell in which particles are places in a regular or irregular manner



"gas (liquid)"



"solid - crystal"

Atomistic description

Atomistic description

Back to the application of diffusion problem…

- Atomistic description provides alternative way to predict *D*
- Simple solve equation of motion
- Follow the trajectory of an atom
- Relate the average distance as function of time from initial point to diffusivity

- **Goal: Calculate how particles move "randomly", away from initial position**

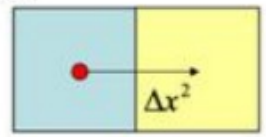
https://physics.weber.edu/schroeder/md/

Link atomistic trajectory with the diffusion constant (1D)

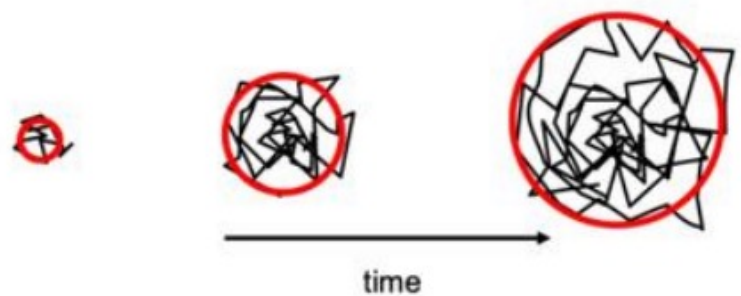
Link atomistic trajectory with diffusion constant (1D)

Diffusion constant relates to the "ability" of a particle to move a distance Δx^2 (from left to right) over a time Δt

$$D = p\frac{\Delta x^2}{\Delta t}$$



Idea – Use MD simulation to measure square of displacement from initial position of particles, $\Delta r^2(t)$:



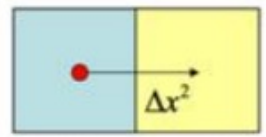
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Link atomistic trajectory with the diffusion constant (1D)

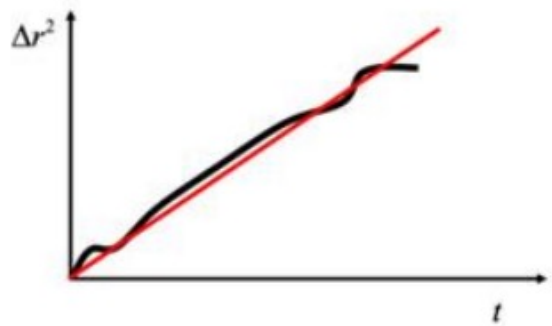
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MD simulation: Measure square of displacement from initial position of particles, $\Delta r^2(t)$:

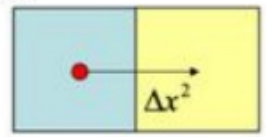


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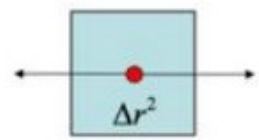
$$D = p\frac{\Delta x^2}{\Delta t}$$



MD simulation: Measure square of displacement from initial position of particles, $\Delta r^2(t)$ and not $\Delta x^2(t)$ ....

Replace

$$D = p\frac{\Delta x^2}{\Delta t} \longrightarrow D = \frac{1}{2}\frac{\Delta r^2}{\Delta t}$$



Factor 1/2 = no directionality in (equal probability to move forth or back)

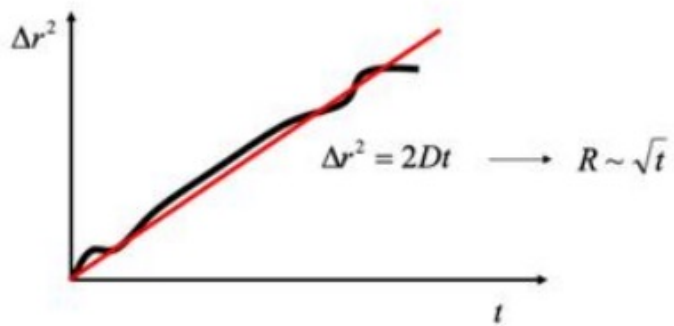
64

Link atomistic trajectory with the diffusion constant (1D)

Link atomistic trajectory with diffusion constant (1D)

MD simulation: Measure square of displacement from initial position of particles, $\Delta r^2(t)$:

$$D = \frac{\Delta r^2}{2t}$$



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Link atomistic trajectory with the diffusion constant (2D/3D)

Link atomistic trajectory with diffusion constant (2D/3D)

$$D = p\frac{\Delta x^2}{\Delta t}$$

Higher dimensions

$$D = \frac{1}{2}\frac{1}{d}\frac{\Delta r^2}{\Delta t}$$

Factor 1/2 = no directionality in (forth/back)

Factor d = 1, 2, or 3 due to 1D, 2D, 3D (dimensionality)

Since:

$$2dD\Delta t \sim \Delta r^2$$

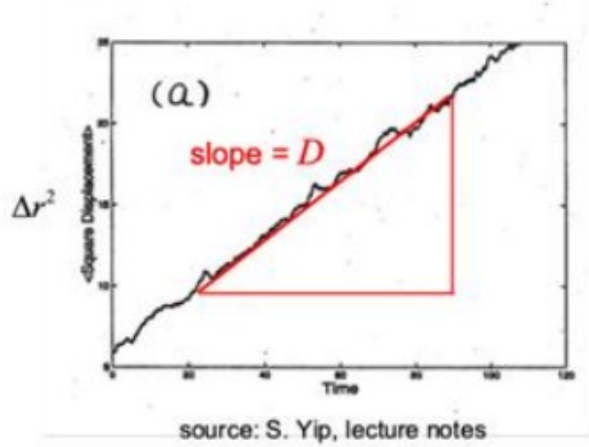
$$\boxed{2dD\Delta t + C = \Delta r^2}$$

C = constant (does not affect D)

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Example: MD simulation

Example: MD simulation



source: S. Yip, lecture notes

$$D = \frac{1}{2d} \lim_{t \to \infty} \frac{\mathrm{d}}{\mathrm{d}t} \left(\Delta r^2 \right)$$

↑

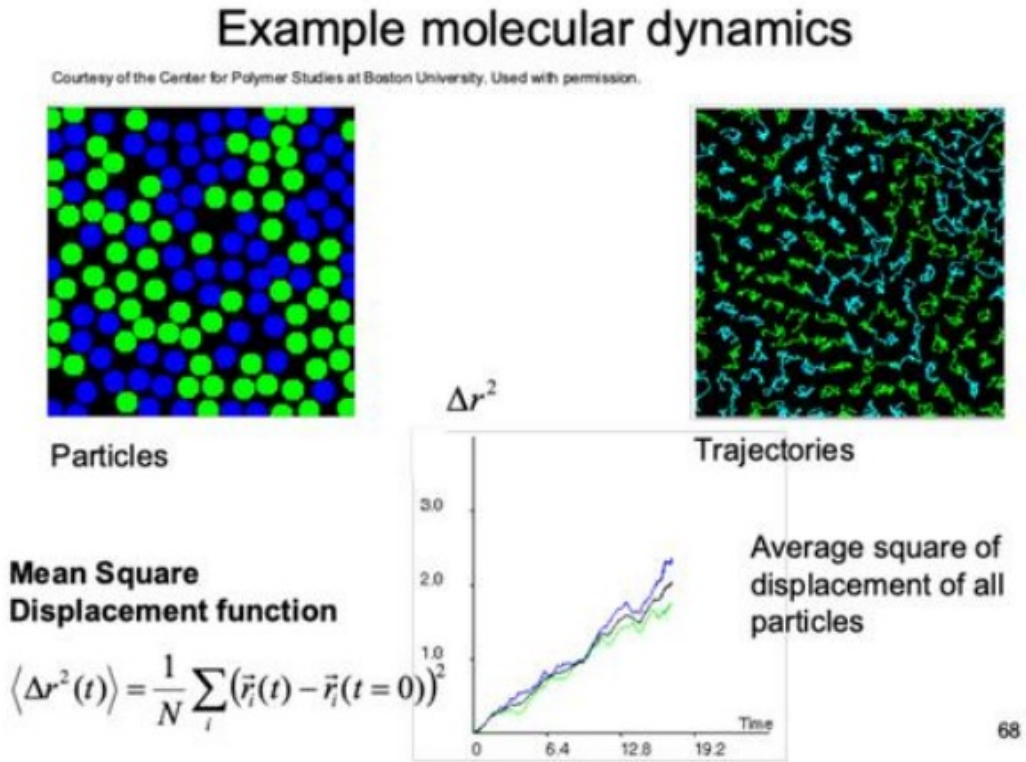
1D=1, 2D=2, 3D=3

$$D = \frac{1}{2d} \lim_{t \to \infty} \frac{\mathrm{d}}{\mathrm{d}t} \left\langle \Delta r^2 \right\rangle$$

$\langle .. \rangle$ = average over all particles

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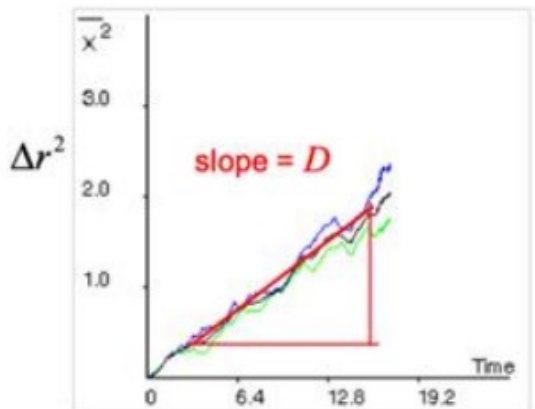
Example molecular dynamics



Example calculation of the diffusion coefficient

Example calculation of diffusion coefficient

$$\left\langle \Delta r^2(t) \right\rangle = \frac{1}{N} \sum_i \underbrace{\left(r_i(t) \right.}_{\text{Position of atom } i \text{ at time t}} - \underbrace{\left. r_i(t=0) \right)^2}_{\text{Position of atom } i \text{ at time t=0}}$$



$$D = \frac{1}{2d} \lim_{t \to \infty} \frac{\mathrm{d}}{\mathrm{d}t} \left\langle \Delta r^2 \right\rangle$$

1D=1, 2D=2, 3D=3

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Summary

Summary

- Molecular dynamics provides a powerful approach to relate the diffusion constant that appears in continuum models to atomistic trajectories

- Outlines multi-scale approach: Feed parameters from atomistic simulations to continuum models

